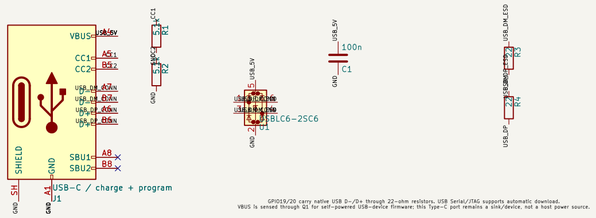
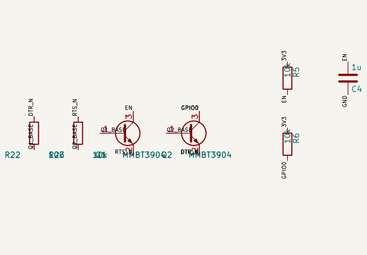
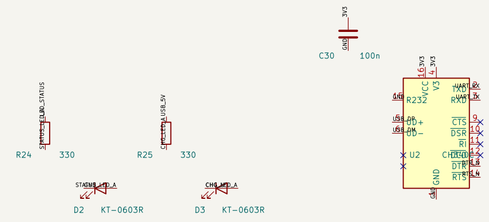
1-2 USB-C INPUT, CC SINK, AND ESD

3-4 CH340C USB-UART / AUTO PROGRAM



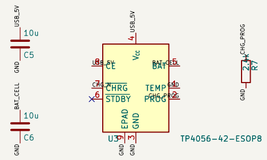
GPIO19/20 carry native USB D-/D+ through 22-ohm resistors. USB Serial/JTAG supports automatic download.
VBUS is sensed through Q1 for self-powered USB-device firmware; this Type-C port remains a sink/device, not a host power source.

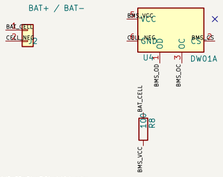




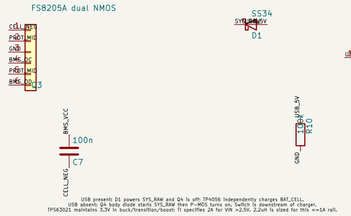
5 TP4056 CHARGER (~500 mA)

6 CELL PROTECTION (DW01A + FS8205A)7-10 LOAD SHARE, PTC, SWITCH, 3.3 V BUCK-BOOST





Overcurrent estimate: 150mV typical / (2×25mΩ) ≈3.0A at high gate;
150mV / (2×35mΩ) ≈2.1A at low gate. Tolerance/temp range is broader.

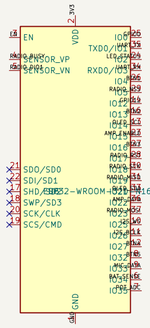


USB presents D1 powers SYS_RAW and Q4 is off: TP4056 independently charges BAT_CELL.
USB absent: Q4 body diode starts SYS_RAW then P-MOS turns on. Switch is downstream of charger.
TPS63021 maintains 3.3V in buck/transition/boost: TI specifies 2A for VIN >2.5V, 2.2uH is sized for this <=1A rail.

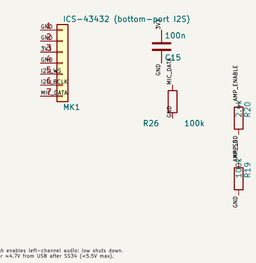


11-12 ESP32-WROOM MODULE, ADC, AND SX1262 RADIO 13-15 I2S MICROPHONE, CLASS-D AMP, SPEAKER

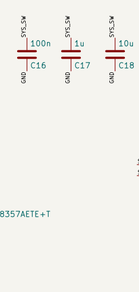
16 COMMERCIAL SSD1306 I2C OLED MODULE HEADER



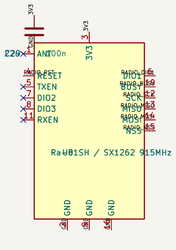




GAIN_SLOT tied directly to GND = +12dB. GPIO4 high enables left-channel audio; low shuts down.
MAX98357A new supply sees +3.0…4.2V battery or +4.7V from USB after SS3A (+5.0V max).



Four through-hole connections only: pin 1 GND, pin 2 3V3, pin 3 SCL, pin 4 SDA.
The commercial module supplies its own SSD1306 charge-pump parts; no raw SYS_5V reaches OLED logic.



17-20 POT, LED, 5× TWO-PIN BUTTONS, TEST PADS



POT: POT_ADC uses ADC1 GPIO2. 100k/100k gives 4.2V -> 2.1V. 21uA drain.
Fine and orderly shutdown about 3.0–3.1V loaded. EW01 cutoff is backup only.

Four through-hole connections only: pin 1 GND, pin 2 3V3, pin 3 SCL, pin 4 SDA.
The commercial module supplies its own SSD1306 charge-pump parts; no raw SYS_5V reaches OLED logic.



Buttons: ESP32-S3 GPIO4/5/6/7/15 -> switch -> GND: firmware enables internal pull-ups.
There are exactly 5 button holes, 5 pot holes, 2 LED holes, 4 OLED holes, and 2 power-switch holes.



See JLC_PART_RESEARCH.md, PIN_ASSIGNMENTS.md, and LAYOUT_NOTES.md.
Power switch must be ON for USB programming; charging works with switch OFF.
TPS63021 fixed buck-boost; 2.2uH; 10uF input; 3x22uF output.
1S Li-ion: WROOM + CH340C USB-UART; SX1262 915MHz; synchronous 3.3V logic

Custom design

Sheet: /
File: walkiepcb.kicad_sch

Title: ESP32 Walkie-Talkie — Economic PCBA schematic

Size: A2	Date: 2026-08-24	**Rev: C**
KiCad E.D.A. 10.0.4		Id: 1/1